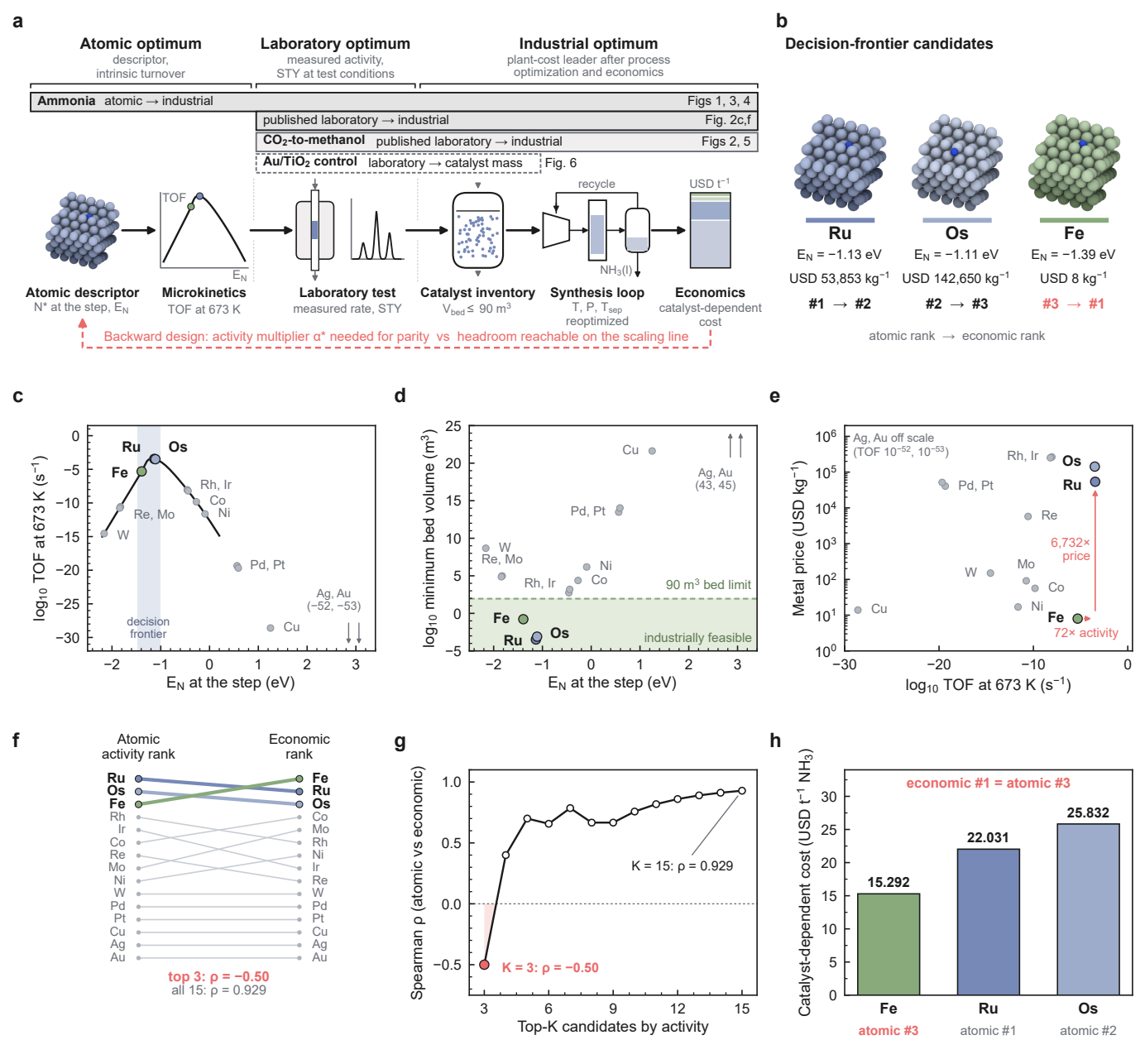

a
Atomic optimum
descriptor, intrinsic turnover
Laboratory optimum
measured activity, STY at test conditions
Industrial optimum
plant-cost leader after process optimization and economics
Ammonia atomic → industrial
Figs 1, 3, 4
published laboratory → industrial
Fig. 2c,f
CO2-to-methanol published laboratory → industrial
Figs 2, 5
Au/TiO2 control laboratory → catalyst mass
Fig. 6
TOF
EN
recycle
NH3(l)
USD t−1
Atomic descriptor
N* at the step, EN
Microkinetics
TOF at 673 K
Laboratory test
measured rate, STY
Catalyst inventory
Vbed ≤ 90 m3
Synthesis loop
T, P, Tsep reoptimized
Economics
catalyst-dependent cost
Backward design: activity multiplier α* needed for parity vs headroom reachable on the scaling line
b
Decision-frontier candidates
Ru
EN = −1.13 eV
USD 53,853 kg−1
#1 → #2
Os
EN = −1.11 eV
USD 142,650 kg−1
#2 → #3
Fe
EN = −1.39 eV
USD 8 kg−1
#3 → #1
atomic rank → economic rank
c
log10 TOF at 673 K (s−1)
EN at the step (eV)
Ru
Os
Fe
Rh, Ir
Co
Ni
Re, Mo
W
Pd, Pt
Ag, Au (−52, −53)
Cu
decision frontier
d
log10 minimum bed volume (m3)
EN at the step (eV)
Cu
Ag, Au (43, 45)
Pd, Pt
W
Re, Mo
Ni
Rh, Ir
Co
90 m3 bed limit
Fe
Ru
Os
industrially feasible
e
Metal price (USD kg−1)
log10 TOF at 673 K (s−1)
Ag, Au off scale (TOF 10−52, 10−53)
Rh, Ir
Os
Ru
Pd, Pt
Re
6,732× price
W
Mo
Co
Cu
Ni
Fe
72× activity
f
Atomic activity rank
Economic rank
Ru
Os
Fe
Rh
Ir
Co
Re
Mo
Ni
W
Pd
Pt
Cu
Ag
Au
Fe
Ru
Os
Co
Mo
Rh
Ni
Ir
Re
W
Pd
Pt
Cu
Ag
Au
top 3: ρ = −0.50
all 15: ρ = 0.929
g
Spearman ρ (atomic vs economic)
Top-K candidates by activity
K = 15: ρ = 0.929
K = 3: ρ = −0.50
h
Catalyst-dependent cost (USD t−1 NH3)
economic #1 = atomic #3
15.292
22.031
25.832
Fe
atomic #3
Ru
atomic #1
Os
atomic #2